Sriram Rajagopalan

S.Rajagopalan@lse.ac.uk ◇ LinkedIn ◇ Website ◇ London, United Kingdom

EDUCATION

London School of Economics and Political Science London, UK
MSc Economics *Aug 2026 – Jun 2028*

Madras School of Economics Chennai, India
B.A Economics (Honours) *Aug 2023 – May 2026*

- CGPA - 9.21
- Relevant Modules - Calculus 1 and 2, Mathematical Statistics, Macroeconomics, Microeconomics, Time Series, Econometrics, Differential Equations, Stochastic Processes, and Real Analysis.

TEACHING EXPERIENCE

Madras School of Economics Chennai, India
Teaching Assistant

Calculus 1 *July 2024 – Nov 2024*
- Facilitated weekly problem-solving sessions on functions, limits, continuity, differentiability, integration, and the Fundamental Theorem of Calculus, clarifying student doubts.

Mathematical Statistics *July 2025 – Nov 2025*
- Conducted and graded weekly tests on probability distributions, point and interval estimation, and hypothesis testing; guided students through problem sets and doubts.

Intermediate Macroeconomics *Jan 2026 – May 2026*
- Lectured on theories such as rational and adaptive expectations; solved weekly problem sets ranging from differential equations to endogenous growth theory.

QUANTITATIVE RESEARCH PROJECTS

Hamilton's Regime Switching Models *2025*
- Authored a rigorous exposition of Markov-switching time-series models, deriving results from Neftçi (1984, *JPE*) and Goldfeld-Quandt (1973, *J. Econometrics*) through to Hamilton (1989).
- Formulated the finite-state Markov structure, the AR(1) representation of the latent regime, and the nonlinear filter and smoother.
- Coded the model in Python and recovered smoothed regime probabilities for US Real GNP, with recession dates closely tracking the NBER chronology.

Empirical Validity of the Phillips Curve in India *2026*
- Tested the inflation-unemployment relationship using national annual time-series and an unbalanced state-level panel (rural and urban, 2017–23).
- Conducted Breusch–Pagan LM and F-tests (both indicating no significant panel effects); estimated pooled OLS with year dummies and state-clustered standard errors, plus an expectations-augmented (lagged-inflation) specification.
- Determined that the unemployment coefficient remained positive and insignificant- **no evidence of a Phillips curve**; inflation instead driven by supply-side (year) shocks and significant inflation persistence (lagged-inflation coefficient 0.27, $p = 0.01$).

Modelling US Recession Arrivals as a Poisson Process *2024 – 2026*
- Co-authored an article in the college magazine testing a homogeneous Poisson model of post-war US recessions using quarterly FRED indicators.
- Evaluated fit via a Chi-Square goodness-of-fit test, a two-sample runs test, and ACF diagnostics.
- Programmed percentile, pivot, bootstrap-t, and parametric bootstrap confidence intervals in Python for the mean, variance, and variance-to-mean ratio.
- Detected sensitivity of arrival counts to the counting convention, motivating a renewal-process extension.

SKILLS

Programming Languages: Python (NumPy, Pandas, Matplotlib, statsmodels), Stata, and LaTeX.
Other Academic Interests: Complex Analysis and Political Philosophy.